Saptarshi Manna

Software Engineering Intern | Full Stack Developer | IT Undergraduate
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Education

Jadavpur University **Nov 2024 – 2028 (Expected)**
Bachelor of Engineering in Information Technology (B.E, IT) | CGPA: 7.5+/10 *Kolkata, West Bengal, India*

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Object-Oriented Systems (OOS), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks, Software Engineering

Technical Skills

- **Languages:** C++, C, Java, Python, JavaScript (ES6+), TypeScript, SQL
- **Frontend Development:** React.js, Next.js, Vite, Tailwind CSS, TanStack Query, HTML5, CSS3, DaisyUI
- **Backend & Cloud:** Node.js, Express.js, RESTful APIs, WebSockets (Socket.IO), Inngest, Redis Docker, AWS, Vercel, Render
- **Databases & Security:** MongoDB, Mongoose, PostgreSQL, MySQL, JWT Authentication & Authorization, Arcjet (Bot Defense & Rate Limiting), Bcrypt
- **Developer Tools & Core CS:** Git, GitHub, Linux/Bash, Postman, Data Structures & Algorithms, Dynamic Programming, OOP, OOS, CN, DBMS, OS

Projects

MPLAD Risk Monitor | *React, GeoJSON, Git, AWS, Vercel, Render* **(Smart India Hackathon)** **Sep 2026**
- Engineered responsive **React + Vite** frontend with **interactive GeoJSON risk maps** to monitor constituency-level public fund utilization and detect expenditure anomalies.
- Developed distinct **role-based portals (District Authority & MP)** with dynamic metric cards, review pipelines, and an audit document management dashboard.
- Integrated an **AI Assistant Chatbot** consuming **FastAPI** REST endpoints and RAG hybrid retrieval to answer audit queries from official scheme guidelines.
- Orchestrated multi-service deployment across **Vercel, Render, and AWS**; managed Git branching workflows and PR code reviews for a **6-member team**.

Coding Interview Platform | *React, Node.js, Monaco Editor, Judge0, Stream SDK, Clerk, Inngest* **Jun 2026**
- Architected a real-time collaborative coding environment pairing **Monaco Editor** code sync with **Stream Video SDK** video calling and persistent live chat.
- Integrated **Judge0 API** for sandboxed code execution across C++, Java, Python, and JS, evaluating test cases with sub-second feedback on stdout, runtime, and errors.
- Enforced **Role-Based Access Control (RBAC)** and route guards using **Clerk Auth**, with session locking and participant management controls.
- Streamlined async jobs via **Inngest background queues** and reduced redundant API calls by leveraging **TanStack Query** for state caching and invalidation.

Real-Time Messaging Platform | *MERN Stack, Socket.IO, JWT, Arcjet, Cloudinary, MongoDB* **May 2026**
- Engineered a low-latency chat platform with **Socket.IO (WebSockets)** on Express, supporting custom event routing, dynamic rooms, and real-time presence tracking.
- Hardened security by embedding **Arcjet rate limiting and bot defense middleware**, paired with stateless **JWT in HttpOnly cookies** and Bcrypt encryption.
- Designed indexed **MongoDB/Mongoose** schemas optimized for chat history pagination and integrated **Cloudinary API** for CDN-delivered media uploads.
- Engineered socket handshake authentication and connection cleanup to eliminate memory leaks; deployed resilient builds on **Vercel** and **Render**.

Achievements & Competitive Programming

- **LeetCode:** Solved **250+ problems** with a contest rating of **1752** (Top 10.76% globally among 880,000+ competitors) | **saptarshi_codes**
- **Codeforces:** Achieved **Pupil** rank with a peak rating of **1383** (Current: 1324) | **Saptarshi_codes**
- **Smart India Hackathon (SIH):** Selected as Team Member/Frontend Lead for building an AI-powered public fund risk monitoring system.
- **Adobe University Hackathon 2026:** Round 2 Qualifier (Advanced past the Round 1 nationwide online coding and assessment stage).
- **WBJEE 2024:** Secured **State Rank 505** among **140,000+ candidates (Top 0.36%)** in West Bengal Joint Entrance Examination, gaining admission to Information Technology at Jadavpur University.